The Fundamental Unit of Life

The big idea

Every living thing, from a bacterium to a banyan tree, is made of **cells**. A cell is the basic structural and functional unit of life: it takes in food, releases energy, removes waste and reproduces.

Discovery of the cell

- **Robert Hooke (1665)** saw tiny boxes in cork and named them cells.
- **Leeuwenhoek (1674)** first saw free-living cells in pond water.
- **Robert Brown (1831)** discovered the nucleus.
- **Schleiden and Schwann (1838–39)**: all plants and animals are made of cells (**cell theory**). **Virchow (1855)**: all cells arise from pre-existing cells.

Organisms can be **unicellular** (Amoeba, Paramecium, bacteria) or **multicellular** (humans, plants). Cells vary in shape and size; nerve cells are long, red blood cells are disc-shaped.

Plasma membrane and cell wall

PLASMA MEMBRANE

The outer covering of every cell, made of lipids and proteins. It is **selectively permeable**: it lets some substances in and out but not others.
Diffusion: movement from high to low concentration (O_2, CO_2).
Osmosis: movement of **water** through a selectively permeable membrane from a dilute to a concentrated solution.

In a **hypotonic** solution a cell swells (water enters); in a **hypertonic** solution it shrinks. A plant cell losing water shrinks away from its wall: **plasmolysis**.

Plant cells (and fungi) also have a **cell wall** made of **cellulose**, which gives strength and lets them survive in dilute surroundings without bursting.

Nucleus

Controls the cell's activities. It contains **chromosomes** made of **DNA** (genetic information) and proteins. Bacteria have no well-defined nucleus; their genetic material lies in a region called the **nucleoid**. These are **prokaryotes**; cells with a membrane-bound nucleus are **eukaryotes**.

Cell organelles

Organelle	Function
Endoplasmic reticulum (rough, with ribosomes / smooth)	Transport channels; RER makes proteins, SER makes lipids and detoxifies
Golgi apparatus	Packages and dispatches materials; forms lysosomes
Lysosomes	Digest waste and worn-out parts: the cell's **"suicide bags"**
Mitochondria	Release energy as ATP: the **powerhouse of the cell**; have their own DNA and ribosomes
Plastids (chloroplasts)	Photosynthesis (in plant cells)
Vacuoles	Storage; large central vacuole in plant cells keeps them turgid
Ribosomes	Protein synthesis

PLANT VS ANIMAL CELLS

Plant cells: cell wall, chloroplasts, large central vacuole.
Animal cells: no cell wall, no chloroplasts, small vacuoles, centrioles.

Cell division

- **Mitosis**: for growth and repair; one cell makes two identical daughter cells with the same number of chromosomes.
- **Meiosis**: in reproductive organs to form gametes; four cells with **half** the number of chromosomes.

Quick recap

- The cell is the unit of life; cell theory by Schleiden, Schwann and Virchow.
- Plasma membrane is selectively permeable; osmosis moves water.
- Know each organelle's role.
- Mitosis for growth, meiosis for gametes.

Practice questions

1. Who discovered cells in cork?
   (a) Robert Hooke (b) Leeuwenhoek (c) Robert Brown (d) Virchow
2. Which organelle is called the powerhouse of the cell?
   (a) Nucleus (b) Golgi apparatus (c) Mitochondria (d) Lysosome
3. Movement of water across a selectively permeable membrane is
   (a) diffusion (b) osmosis (c) plasmolysis (d) endocytosis
4. Lysosomes are called suicide bags because they
   (a) can digest the cell's own material (b) make proteins (c) store water (d) release energy
5. The cell wall of plants is made of
   (a) protein (b) lipid (c) cellulose (d) chitin

Answers: 1. a 2. c 3. b 4. a 5. c